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A MIRI Imaging Search for Outer Companions to WD 1856 b

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ABSTRACT

WD 1856+534 b is the first known planet transiting a white dwarf. However, it could not have survived its host star's evolution off the main sequence at its current close-in orbit (P = 1.4 days). One proposed explanation is that WD 1856 b began at a wider separation, but underwent planet-planet scattering and high-eccentricity migration to reach its present position. To search for potential perturbing companions, we use two epochs of JWST MIRI imaging observations to measure the proper motions of all nearby sources. While we do not identify any co-moving planetary companions, we are able to place the first strong constraints on the presence of any distant outer planets—we rule out planets >7.6−8.7 M_{Jup} at >22 au, >5.3−6.1 M_{Jup} at 50−110 au, and >4.5−5.3 M_{Jup} between >110 au and WD 1856's Hill sphere radius. The provided ranges correspond to lower and upper limits on the system's age (7.4−10 Gyr). Additionally, we leverage NIRSpec PRISM and qCMOS observations to provide updated transit timing variation (TTV) constraints. For edge-on and circular orbits, we rule out any companions >2.3 M_{Jup} at separations <5 au (P<5200 days). Our combined constraints rule out planets more massive than WD 1856 b across much of the searched parameter space. If no companion is found within the remaining allowed region, it suggests that either no additional planets are present in the system and WD 1856b migrated via a non-scattering mechanism, or any perturbing planet has a lower mass than WD 1856 b and was likely ejected from the system.

Keywords: planetary systems

1. INTRODUCTION

The overwhelming majority of known exoplanets orbit main sequence stars. However, there is abundant evidence that planetary systems can survive around post-MS stars, including white dwarf photospheric metal pollution (Jura 2003; Farihi et al. 2009; Zuckerman et al. 2010; Koester et al. 2014) and several white dwarf systems with transiting debris clouds, believed to be the result of disintegrating rocky bodies (e.g., WD 1145+017, Vanderburg et al. 2015; Gänsicke et al. 2016). Intact planets and planet candidates have also been found orbiting WDs (Thorsett et al. 1993; Luhman et al. 2011; Gänsicke et al. 2019; Limbach et al. 2024a). Notably, the only known planet transiting a white dwarf is WD 1856+534 b (hereafter WD 1856 b, Vanderburg et al. 2020) .

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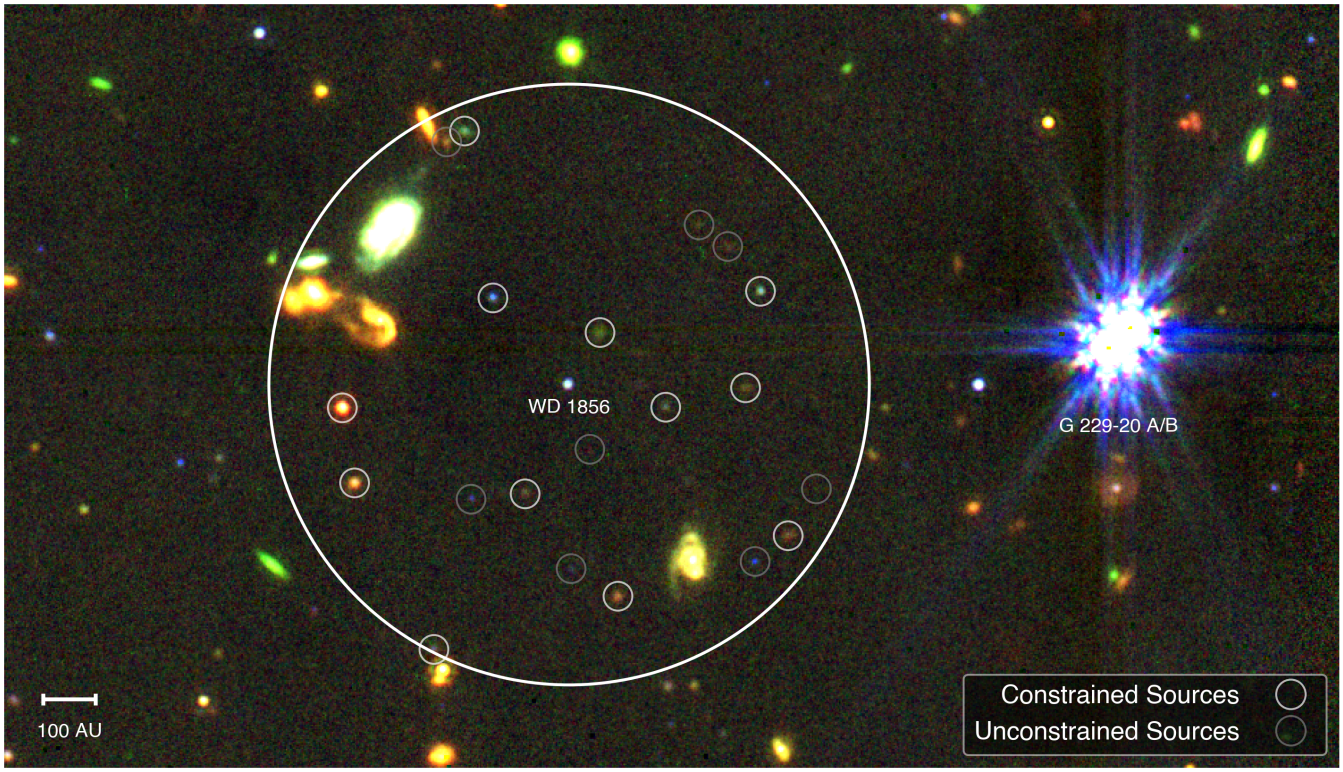


Figure 1. False-color image of the WD 1856 system made using three JWST MIRI filters: F770W, F1500W, and F1800W. We label both WD 1856 and its companion M dwarf binary, G 229-20 A/B. We denote WD 1856's Hill sphere ($R_{\rm HS} = 570^{+250}_{-170}$ au, Vanderburg et al. 2020; Limbach et al. 2025) with a large white circle. To search for additional planets in the system, we measure the proper motions of all sources within this region. Sources with well-constrained motions ($\sigma < 0.5$ pixels) are marked in white, and sources with poorly constrained motions are marked in gray. We describe the process of identifying nearby sources and measuring their proper motions in §3.2.

Discovered in 2020 with data from NASA's Transiting Exoplanet Survey Satellite (*TESS*), WD 1856 b is a 0.946±0.017 $R_{\rm Jup}$ planet orbiting its host star at an orbital separation of 0.02 au (Vanderburg et al. 2020; Limbach et al. 2025). Using data from the JWST Mid-Infrared Instrument (MIRI), Limbach et al. (2025) constrained the planet's mass to be $5.2^{+0.7}_{-0.8}$ $M_{\rm Jup}$ assuming insolation has a negligible effect on the planet's thermal evolution, and estimated its effective temperature to be 184±8 K, making it the coldest planet with directly measured emission. However, its close-in orbit raises important questions about its dynamical history. Given that stars expand to ∼1-2 au as they evolve on the red giant branch and asymptotic giant branch, they are expected to carve out a so-called "forbidden zone" that is devoid of planets. WD 1856 b is the first intact planet confirmed to orbit within this forbidden zone.

Two primary scenarios have been proposed to explain this close-in orbit. First, WD 1856 b may have undergone common envelope evolution (Chamandy et al. 2021; Lagos et al. 2021; Merlov et al. 2021), a process in which the planet is engulfed as the star evolves off the main sequence. The planet then rapidly spirals inwards. If sufficient energy is injected to expel the stellar envelope, then the planet may survive at a new, closer-in orbit. The second explanation is that the planet underwent high-eccentricity migration after the star had evolved to its final state as a white dwarf. In this scenario, the planet starts at a wider orbit around the white dwarf star and is driven to high eccentricities. This could be accomplished, for example, by the gravitationally bound M dwarf binary G 229-20 A/B, which has a projected separation of ∼1000 au from the primary white dwarf and may drive Kozai-Lidov cycles in WD 1856 b's orbit (Muñoz & Petrovich 2020; O'Connor et al. 2021; Stephan et al. 2021). Alternatively, a secondary planetary companion could excite WD 1856 b to high eccentricities via planet-planet scattering (Maldonado et al. 2021; O'Connor et al. 2022). In either scenario, WD 1856 b would undergo high eccentricity migration and tidal circularization to reach its current orbit (Veras & Fuller 2019; Veras et al. 2019).

In this work, we investigate the possibility that a perturbing planetary companion led to WD 1856 b's migration. Kubiak et al. (2023) previously placed constraints on any companions by searching for changes to WD 1856 b's transit times, called transit timing variations (TTVs). Using data collected by Mallonn (2020), Vanderburg et al. (2020), Alonso et al. (2021), and Xu et al. (2021), in addition to data collected using the 16 inch privately-owned telescope at Hereford Arizona Observatory (HAO), they constrain any companions to be less than 2 $M_{\rm Jup}$ for orbital periods up to 500 days, and less than 10 $M_{\rm Jup}$ for orbital periods up to 1000 days. Gendreau-

Distler et al. (2025) have since calculated updated constraints using an additional 20 transits, ruling out planets less than 4.1 M_{Jup} for orbital periods less than 1500 days. We build on this earlier work by searching for a planetary companion with a combination of TTVs and direct imaging.

Direct imaging has been used to discover dozens of planets (e.g., Chauvin et al. 2005; Marois et al. 2008; Lagrange et al. 2010; Macintosh et al. 2015; Currie et al. 2023), with a primary focus on observing massive planets around young stars. However, imaging is also a particularly promising technique for searching for planets around white dwarf stars, as white dwarfs are significantly fainter than main sequence stars. In particular, emission from high-mass planetary companions can be comparable to emission from the white dwarf host at mid-infrared wavelengths. Direct imaging has already been successfully used to discover a planetary-mass object around WD 0806-661 (Rodriguez et al. 2011), and several JWST programs (Mullally et al. 2021; Poulsen et al. 2023; Limbach et al. 2024b; Vanderburg et al. 2025; Sallum et al. 2025) are dedicated to searching for planets around white dwarfs with a combination of direct imaging and the infrared excess method (Stevenson & Space Telescopes Advanced Research Group on the Atmospheres of Transiting Exoplanets 2020; Limbach et al. 2022). While these programs have identified several planet candidates (Mullally et al. 2024; Albert et al. 2025; Limbach et al. 2024a), follow-up observations are needed to confirm whether they are true planets. Close-in planets discovered with the infrared excess technique require spectroscopic observations to rule out a disk, while more distant planets discovered with imaging require a second epoch of imaging observations to rule out spurious background sources. This second epoch provides a constraint on the candidate's proper motion, and reveals whether the source is co-moving with, and therefore gravitationally bound to, the white dwarf. Imaging follow-up has already ruled out two candidates (Mullally et al. 2025).

In this work, we use two epochs of JWST MIRI imaging to search for co-moving planets in the WD 1856 system and to place the first strong constraints on any distant, outer companions. Additionally, we provide updated TTV constraints using two additional transit observations. The paper is organized as follows: we describe the imaging data and transit observation in §2. We then detail our search for outer companions in §3 and place limits on the presence of any existing companions in §4. We discuss the implications for the dynamical history of WD 1856 b and prospects for future companion searches in §5, before concluding in §6.

2. OBSERVATIONS AND DATA REDUCTION

2.1. *JWST MIRI Imaging*

The WD 1856 system was imaged as part of the JWST Cycle 3 program "Probing the Dynamical History and the Mid-IR SED of WD 1856b" (GO #5204, Limbach et al. 2024c). This program had two primary objectives. First, it measured thermal emission from WD 1856 b. Because the planet is close-in and unresolved, this required measuring the combined flux from the white dwarf and planet, and searching for a mid-infrared excess. These results constrained the planet's temperature (184±8 K) and mass ($5.2^{+0.7}_{-0.8}$ M_{Jup}, see Table 1 for a full list of system properties), thus confirming that WD 1856 b is a planetary mass object (Limbach et al. 2025). The second goal of this program was to search for additional planets in the WD 1856 system using direct imaging. With the second epoch of observations scheduled nearly a year after the first epoch, we had a sufficient baseline to measure the proper motions of all nearby sources and to identify any sources co-moving with the white dwarf. If a planet is responsible for scattering WD 1856 b from its original position, it is likely also high-mass. With a predicted sensitivity to planets down to 5 M_{Jup}, this program was well positioned to identify any such perturbing companions.

We observed the WD 1856 system with JWST MIRI on 2024 July 29 and 2025 July 28. Due to its high proper motion, the WD 1856 system is expected to move 246 mas over the program's baseline of nearly one year. The first epoch of data imaged the system with seven filters: F560W, F770W, F1000W, F1130W, F1280W, F1500W, and F1800W (with central wavelengths of 5.6, 7.7, 10.0, 11.3, 12.8, 15.0, and 18.0 μm, respectively). The second epoch of data imaged the system in only the two reddest filters, F1500W and F1800W, as any planets are expected to be brightest at these wavelengths. We therefore focus on these two filters in our search for planetary companions. For the F1500W filter, we used 40 groups per integration and four integrations per observing epoch, resulting in a total exposure time of 7.4 minutes per epoch. For the F1800W filter, we used 40 groups per integration and 12 integrations per observing epoch, resulting

Table 1. WD 1856 System Properties.

Parameter	Value	Source
	Stellar Properties	
RA	18:57:39.34	*Gaia**
Dec	+53:30:33.3	*Gaia**
d (pc)	24.754 ± 0.044	*Gaia**
$\mu_\alpha \cos\delta$ (mas yr^{-1})	240.759 ± 0.148	*Gaia**
μ_δ (mas yr^{-1})	−52.514 ± 0.143	*Gaia**
M ($M_\odot$)	0.605 ± 0.013	Limbach et al. (2025)
R ($R_\odot$)	0.0121 ± 0.0002	Limbach et al. (2025)
$T_{\rm eff}$ (K)	4920 ± 50	Limbach et al. (2025)
$t_{\rm cool}$ (Gyr)	5.4 ± 0.7	Limbach et al. (2025)
$t_{\rm total}$ (Gyr)	7.4−10	Limbach et al. (2025)
	WD 1856 b Properties	
M (M_{Jup})	$5.2^{+0.7}_{-0.8}$	Limbach et al. (2025)
R (R_{Jup})	0.946 ± 0.017	Limbach et al. (2025)
P (days)	1.407939217 ± 0.000000016	Kubiak et al. (2023)
a (au)	0.02085 ± 0.0014	Kubiak et al. (2023)
$T_{\rm eff}$ (K)	184 ± 8	Limbach et al. (2025)

*Gaia Collaboration et al. (2016a, 2018, 2023)

in a total exposure time of 22.6 minutes per epoch. These integration times were chosen to achieve a signal-to-noise (SNR) ratio of ≈50 on the white dwarf in each filter with the combined epochs. All observations used the fast readout mode and four-point cycling dither, which helps minimize the effects of cosmic ray showers, background noise, and bad pixels. We additionally used the full MIRI imaging array, which has a 74" × 113" field of view (FOV). Given the system's distance of 24.8 pc (Vanderburg et al. 2020), this corresponds to a region of 1835 × 2802 au around the white dwarf.

The MIRI imaging products from the `jwst` pipeline exhibit a non-uniform background that is strongest at the longest wavelengths. Therefore, we reprocess the `jwst` Stage 2 outputs using a custom software package, `Magic`[1], that applies background subtraction following the steps from a STScI JWebbinar tutorial[2]. This method constructs a background model from the multiple dither positions for a given target and produces Stage 2 data products with improved background uniformity. As a final step in the reduction process, we combine the dithers into a single image using the `jwst` Stage 3 pipeline. We show a false-color image of the system in Figure 1.

2.2. *JWST PRISM Spectroscopy*

In addition to searching for companions with direct imaging, we also search for evidence of a companion using transit timing variations (TTVs). To strengthen existing TTV constraints from Kubiak et al. (2023) and Gendreau-Distler et al. (2025), we use an additional transit observation included as part of the JWST Cycle 1 program "Under the Light of a Dead Star: Revealing the Atmospheric Composition of a White Dwarf Planet" (GO #2358, MacDonald et al. 2021). This program observed the system for 2 hours on 2023 April 27 with the Near Infrared Spectrograph (NIRSpec) PRISM in Bright Object Time Series (BOTS) mode. It used 33 groups per integration and 240 integrations per observing epoch. It additionally used the NRSRAPID readout pattern.

MacDonald et al. (2026) reduced the light curve using the Fast InfraRed Exoplanet Fitting Lyghtcurve (`FIREFLy`, e.g., Rustamkulov et al. 2022; Rustamkulov et al. 2023; Sing et al. 2024) pipeline. In stages 1 and 2, they removed the 1/f noise at the group-level, skipped the dark current step, and performed the jump step with a rejection threshold of 20. In stage 3, they performed a custom removal of bad pixels, cosmic rays, and hot pixels, shift-stabilized the images (Rustamkulov et al. 2022; Rustamkulov et al. 2023), and used an aperture of 5.7 pixels to extract the spectra. In stages 4 and 5, they fitted the resulting white light curve. The white light curve was calculated using the flux between 0.55 and 3 μm to avoid contamination by planetary emission. They fitted the light curve using the Python package `batman` (Kreidberg 2015). The period was fixed to the best-fit value from Kubiak et al. (2023), and the quadratic limb-darkening parameters were fixed to the best-fitting values for a white dwarf model, calculated using `ExoTiC-LD` (Grant & Wakeford 2024). They estimated a best-fit transit time of 2460062.00770117±0.0000005, in BJD TDB (private communications).

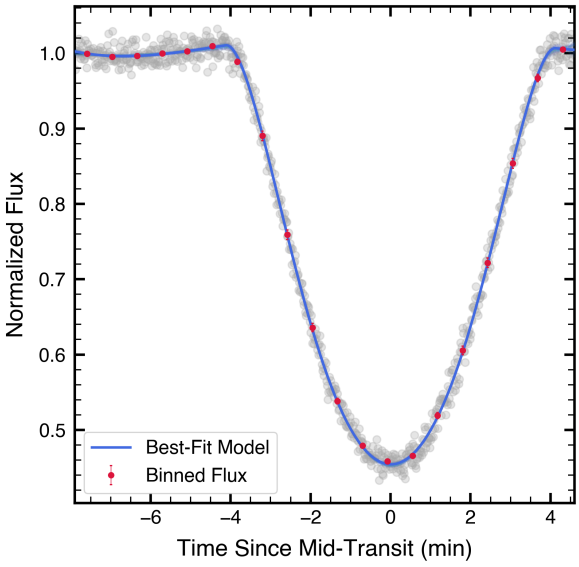


Figure 2. Photometric data from qCMOS photometry. Binned data is shown in red and the best-fit model is shown in blue. We use the best-fit transit time to provide updated TTV constraints on the presence of any existing outer companions.

2.3. *qCMOS Photometry*

We also observe an additional transit of WD 1856 using an early iteration of the proto-Lightspeed instrument (Layden et al. 2026). In this iteration, a Hamamatsu ORCA-Quest 2 (qCMOS) camera was mounted at the prime focus of the Hale Telescope at Palomar Observatory, with no re-imaging optics. We observed the system on 2024 October 26 with an exposure time of 1 s and no filter. We note that we collect a limited post-transit baseline. We used the ultra-quiet readout mode and the camera's photon number resolving option. This option converts the digital output of each pixel to an estimated number of collected photoelectrons. Because the read noise of most of the pixels in the camera is less than 0.3 e^-, this conversion can be done with a reasonably high level of confidence, yielding an even lower effective read noise per pixel (Layden et al. 2026; Teranishi 2012). However, this option has a limited dynamic range (200 e^-) and may yield a slightly deteriorated effective read noise for pixels with original read noise above 0.3 e^-.

We correct for the nonlinear response of this ORCA-Quest 2 camera, which arises from incomplete charge transfer within each pixel, by dividing each recorded number of photoelectrons by the camera's average internal quantum efficiency

[1] https://github.com/kevin218/Magic

[2] https://github.com/spacetelescope/jwebbinar_prep/blob/jwebbinar31/jwebbinar31/miri/Pipeline_demo_subtract_imager_background-platform.ipynb

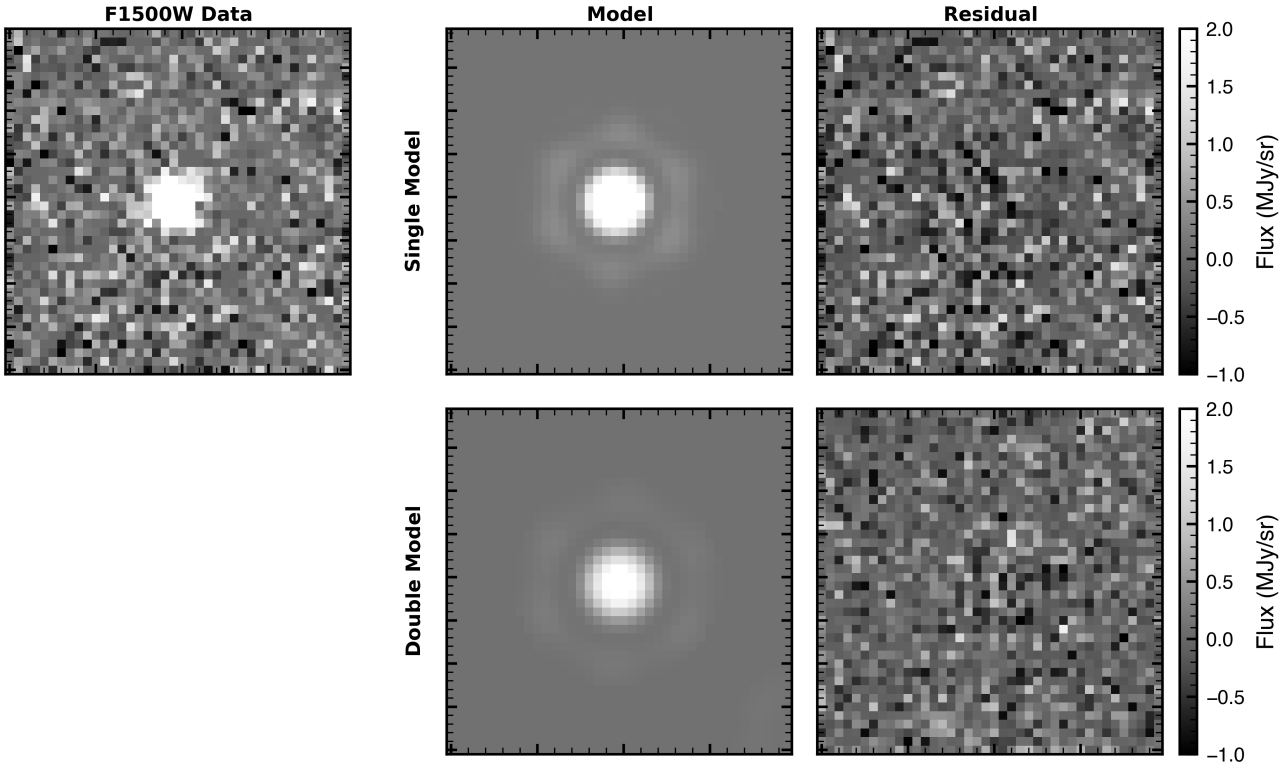


Figure 3. Empirical PSF fits. To search for close-in companions, we simultaneously fit WD 1856 in the four F1500W dithers from both visits. Here we show the data, best-fit models, and residuals for the first dither. We find that the double PSF model has a higher BIC and AIC than the single PSF model. Additionally, the Bayes factor B indicates that there is no evidence the double PSF model is preferred. We therefore conclude that there is no detection of a close-in planet.

at that number of recorded photoelectrons (see Fig. 4c of Layden et al. (2026)). While an individualized pixel-based correction for the photon number resolving option is not yet available for this camera, using the average internal quantum efficiency is sufficient here, where the point-spread function is significantly over-sampled and where most pixels of the target are bright enough to be near the linear regime. We then perform aperture photometry using the Java image processing program AstroImageJ (Collins et al. 2017). We use four nearby comparison stars to calibrate the data. This helps mitigate effects such as changing atmospheric conditions and instrumental systematics.

To measure the transit time, we fit the light curve using the `batman` package. We fit the time of inferior conjunction T_0, planet-to-star radius ratio $R_P/R_\star$, semi-major axis $a/R_\star$, inclination $\cos i$, and quadratic limb-darkening coefficients μ_1 and μ_2. We set Gaussian priors on $R_P/R_\star$, $a/R_\star$, and $\cos i$. The $R_P/R_\star$ and $a/R_\star$ priors are centered on best-fit values from Xu et al. (2021), Kubiak et al. (2023), and Limbach et al. (2025) with standard deviations corresponding to the measured uncertainties. The $\cos i$ prior is similarly centered on the best-fit value found by Vanderburg et al. (2020). Because the tidal circularization timescale for Jupiter-mass planets (~2 Myr, calculated using the formalism introduced by Goldreich (1963) and Goldreich & Soter (1966)) is much less than the estimated age of the system (7.4−10 Gyr, Limbach et al. 2025), we assume an eccentricity of zero. We additionally fit a jitter term, which is added in quadrature to the predicted uncertainties to account for any additional sources of astrophysical or instrumental uncertainties. In total, our likelihood function includes seven free parameters. Before calculating the likelihood at each model iteration, we subtract our transit model from the data and fit a fourth-order polynomial to the residual light curve at each MCMC link. We then add this curve to the transit model. This implicitly includes a correction for and allows us to marginalize over out-of-transit variations in our model without having to explicitly sample those parameters.

We explore the parameter space with a differential evolution Markov Chain Monte Carlo (MCMC) sampler. Sampling is done using the `edmcmc` package (Vanderburg 2021). We evolve 128 chains for 10,000 iterations each, discarding the first 2,000 iterations as burn-in. To test for convergence, we calculate the Gelman-Rubin statistic for each parameter, and find that all values are below 1.01. We estimate a best-fit transit time of 2460609.69607±0.00002, in BJD TDB, consistent with the ephemeris predicted by Kubiak et al. (2023). The data and final transit model are shown in Figure 2. We use the estimated transit times for both the JWST NIRSpec and qCMOS observations to update TTV constraints on the presence of an outer companion, as described in §4.3.

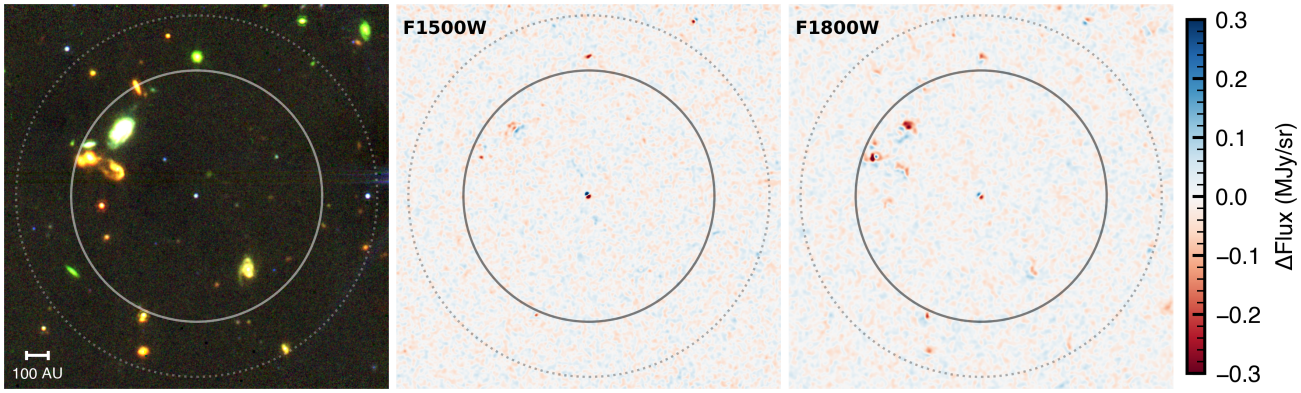


Figure 4. Difference images in both the F1500W and F1800W filters, smoothed with a Gaussian kernel matching the instrument PSF. Distant background objects show negligible apparent motion, and therefore have small residuals. In contrast, nearby and fast-moving objects produce prominent positive and negative residuals. The white dwarf, center, exhibits clear motion. No other sources appear to be co-moving with the white dwarf. The Hill sphere radius is denoted by white and black circles in each image, while its corresponding 1σ upper limit is denoted by white and black dotted lines. *Left:* False-color image. *Middle:* Difference image in the F1500W filter. *Right:* Difference image in the F1800W filter.

3. THE SEARCH FOR PLANETARY COMPANIONS

To identify any planetary companions in the JWST MIRI imaging data, we use a combination of two methods. First, we search for close-in, marginally resolved planets using a point-spread function (PSF) fitting algorithm. We then search for distant, resolved companions using the proper motion technique. We describe each method in more detail below.

3.1. *Searching for Inner Planets with PSF Fitting*

To search for close-in planets at projected separations $\lesssim$30 au ($\lesssim$1.2"), we fit empirical PSF models (e.g., De Furio et al. 2019, 2022, 2023; Mehta et al. 2026). This technique has previously been used to identify the two known Y-Y dwarf binary systems at separations of $\sim 0.5\lambda/D$ with NIRCam in the F150W and F480M filters (Calissendorff et al. 2023) and MIRI in the F1000W and F1280W filters (De Furio et al. 2025). Because white dwarfs, giant planets, and Y-dwarfs are a similar brightness at mid-infrared wavelengths, we expect this method to work equally well in our case. We use super-sampled empirical PSF models from Libralato et al. (2023), which vary as a function of the filter and detector position (e.g., Anderson & King 2000; Anderson 2016). In order to compare these models to the observed PSF, we must convert them to the correct pixel scale. We do this by first performing a bicubic interpolation on each of the four nearest super-sampled PSFs, providing a single pixel value for each of the PSFs. We then perform a linear interpolation on the resulting values, weighted based on each PSF's proximity to the center of the source. This provides a single value for one pixel. We perform these interpolations across the whole cutout where we are making the PSF model.

To find best-fit single and double PSF models, we perform a fit on a 41×41 cutout centered on WD 1856. Because any planet is expected to be brightest in the mid-infrared, we perform a simultaneous fit of both visits in the F1500W filter. We separately fit an x and y coordinate for each dither. Combined, our single PSF model includes a total of 18 free parameters. Our double PSF model includes three additional parameters: the relative x- and y-offsets between the primary and secondary PSFs and the difference in magnitude between the primary and companion. To capture sources on the edge of the cutout, we allow the center position of the secondary PSF to be up to 10 pixels past the edge of the cutout. We sample the parameter space using the Nested Sampling routine PyMultiNest (Feroz et al. 2009; Buchner et al. 2014), following Calissendorff et al. (2023), De Furio et al. (2023), and De Furio et al. (2025). We use 1800 and 2100 live points for the single and double PSF fits, respectively. To improve the sampling of the companion's parameters, we independently fit for companions in each of the four quadrants of the image and select the best-fit companion parameters from these four fits. The best-fit single and double PSF models for the first dither are shown in Figure 3, along with corresponding residuals. We find that the best-fit companion is located 33.8 pixels from the primary PSF, placing it beyond the edge of the cutout. We compare this to our calculated 90% completeness threshold and false positive probability curves in §4.1.

We find no statistical evidence that the double PSF model is a better fit to the data. Both models have a reduced chi-squared statistic of 0.82, though this metric does not fully account for the additional flexibility provided by the extra free parameters. To better compare the two models, we also calculate the Bayesian Information Criteria (BIC, Schwarz 1978) and the Akaike information criterion (AIC, Akaike 1974). In both cases, a smaller value indicates a better model. We find that the double planet model has a higher BIC and AIC, with a ΔBIC of 26.4 and ΔAIC of 6.0. The single PSF model is therefore strongly favored, meaning we do not identify any close-in companions. In addition, we calculate the Bayes factor B, defined as the ratio between the models' marginal likelihoods. With a $\log_{10}B$ value of 0.47, there is no evidence that the double PSF model is preferred (Jeffreys 1939). While we do not identify any close-in planets, we know that there is a planet, WD 1856 b, at just 0.02 au.

However, we do not expect the PSF fitting technique to be sensitive to such a close-in, unresolved planet. We evaluate our sensitivity to planets as a function of planet separation, contrast, and mass in §4.1.

3.2. *Searching for Outer Planets with Proper Motion Measurements*

To search for more distant planets, we look for sources that are co-moving with the white dwarf. To visually inspect the motion of each source, we align the observations from the first and second visit using `Astroalign` (Beroiz et al. 2020), and subtract them to produce a difference image for each filter (see Figure 4). In a difference image, we expect any stationary background objects to have negligible residuals. In contrast, nearby and fast-moving objects, such as WD 1856, exhibit clear motion, as denoted by the prominent positive and negative residuals. We note that there are also residuals around several bright, nearby galaxies due to imperfect subtraction. However, we do not identify any sources co-moving with the white dwarf in the difference images.

To confirm this non-detection, we also measure the proper motion of each nearby source. We identify sources in the F1500W and F1800W bands using the DAOFIND algorithm (Stetson 1987) implemented by the Python package `Photutils` (Bradley et al. 2025). This algorithm convolves the image with a two-dimensional Gaussian kernel and searches for local maxima above a given threshold. In our search, we require sources to have an amplitude greater than 3.5σ. Additionally, we require that all sources are within the white dwarf's Hill sphere ($R_{\rm HS} = 570^{+250}_{-170}$ au, Vanderburg et al. 2020; Limbach et al. 2025), defined as the region within which the white dwarf's gravity dominates. Beyond this region, we expect the nearby M dwarf binary G 229-20 A/B to be gravitationally dominant. Because the Hill sphere radius has large uncertainties, we additionally perform a visual check for companions outside of this region, and do not identify any co-moving sources. After collecting our list of targets, we visually inspect each detection to verify that it is a true source without any clear extended features. Using this method, we identify fourteen sources. However, in combined false-color images, several additional faint sources are visible. We therefore manually add these to our target list, providing an additional six sources. The final set of sources is shown in Figure 1.

For each point source, we simultaneously measure motion and photometry by performing a combined fit on both the first and second visit in the F1500W and F1800W filters. We fit the empirical PSF models described in §3.1 to 21×21 pixel cutouts centered on each source. We perform our fits on the combined Stage 3 data, as we expect the white dwarf and any planets to move by over two pixels over the one year baseline, meaning observational uncertainties will dominate over any systematic errors from image combination. We fit eight free parameters: the x and y coordinates for both visits, and the source and background flux for both filters. As before, we sample the parameter space using PyMultiNest. However, because we fit fewer free parameters, we decrease the number of live points to 800. Using the best-fit coordinates for each visit, we calculate the motion of each source in the x and y direction. We note that three of our sources have nearby companions, making it difficult to fit a single PSF model. For these targets, we simultaneously fit a second PSF, doubling the number of free parameters, and adopt custom cutout sizes to fully capture both sources. Of the 20 total sources, we find that 13 have well-constrained motion, defined as motion measured to within 0.5 pixels (equivalent to 0.055").

Because many of these targets are distant background objects, we expect their motion on the sky to be negligible. This means that the apparent motion of most sources is due primarily to the change in the telescope pointing, including an offset and small rotation. We show the x and y motion of each source, correcting for the telescope pointing, in Figure 5. We only include the 13 well-constrained sources. This eliminates many of the faintest sources, including five of the manually added targets. Properly constraining the proper motion of these faint sources will require additional follow-up imaging with a longer temporal baseline. As shown in Figure 5, the field sources tend to have a motion of ∼0" over the course of the year, consistent with distant background objects. We do not identify any sources co-moving with the white dwarf, denoted by the red cross. We additionally compare the photometry of all sources to the values predicted by blackbody and planet models. Because we measure each source's flux using empirical PSF models, which extend to 12.5 pixels and are therefore expected to only capture ∼85% of a source's light in the F1500W and F1800W filters, we apply an additional aperture correction to estimate the total flux. The blackbody model is calculated for planets between 0.9−2.0 $M_{\rm Jup}$ at varying temperatures, and the planet models are calculated using the BEX cooling tracks (Linder et al. 2019) and ATMO atmospheric models (Chabrier et al. 2023), as implemented in the `MADYS` Python package (Squicciarini & Bonavita 2022). Because many of the unconstrained sources are faint in the F1800W filter, they fall outside the limits of the plot. We note that several background sources have photometry consistent with a planet's, demonstrating that two-color photometry with the F1500W and F1800W filters is insufficient for distinguishing between planets and false positives. Proper motion measurements are therefore critical for identifying true planets. One of these sources has unconstrained motion (with a combined uncertainty of 0.7 pixels). Its measured motion, corrected for changes in telescope pointing, is 0.09±0.05″ and 0.09±0.06″ in the x and y direction, respectively. While this remains inconsistent with the white dwarf's observed motion, further follow-up would help improve this constraint.

4. SENSITIVITY LIMITS

Though we do not identify any additional planets in the WD 1856 system, we can constrain the presence of any existing companions. We first place limits on the sensitivity of our search methods using source injection. We then extend existing TTV constraints from Kubiak et al. (2023) and

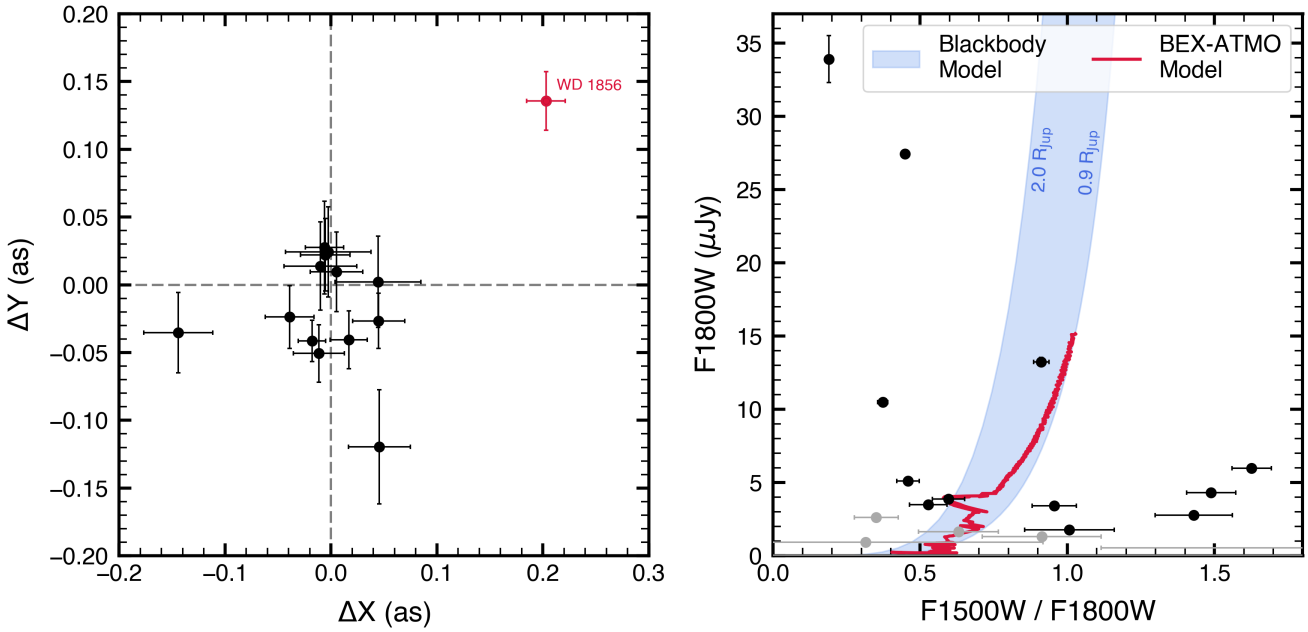


Figure 5. *Left:* Motions of well-constrained sources over one year baseline, accounting for change in telescope pointing. In contrast to WD 1856's large apparent motion (shown in red), we expect distant background objects to exhibit negligible motion. We do not identify any sources co-moving with the white dwarf. *Right:* Photometry of well-constrained (black) and poorly constrained (gray) sources in the F1500W and F1800W filters. For comparison, we show the color range for a blackbody between 0.9–2.0 M_{Jup} in blue, and the color range for a planet modeled using the BEX cooling tracks (Linder et al. 2019) and ATMO atmospheric models (Chabrier et al. 2023) in red. Some background sources have photometry consistent with the theoretical planet values, emphasizing the importance of proper motion measurements in validating planet candidates.

Gendreau-Distler et al. (2025) using our additional transit observations, described in §2.2 and §2.3.

4.1. *PSF Fitting Limits*

To measure the sensitivity of the empirical PSF fitting routine, we generate synthetic binaries using the empirical PSF models described in §3.1. This follows the approach used by e.g., De Furio et al. (2019, 2022) and Mehta et al. (2026). However, because these PSF models are only calculated for separations out to 12 pixels, we generate our synthetic binaries by combining the empirical PSFs with Space Telescope PSFs (STPSFs, Perrin et al. 2012, 2014) for larger separations. We first generate 10,000 binaries with semimajor axes a between 1 and 100 au, and contrasts between 0 and 5 mag. We sample a uniformly on a logarithmic scale, allowing us to better measure the inner limits of our sensitivity. To convert a to an observed separation, we use the Python package `orbitize!` (Blunt et al. 2020), adopting a uniform distribution in sin i, the argument of periastron, and the longitude of the ascending node. We assume that eccentricity follows a Beta distribution with $\alpha = 0.7$ and $\beta = 2.3$, as measured for a sample of directly imaged giant planets (Bowler et al. 2020; Nagpal et al. 2023). To characterize where each planet is in its orbit, we define a parameter τ in terms of the epoch of periastron t_p, a reference epoch $t_{\rm ref}$, and the planet's orbital period P:

$$\tau = \frac{t_p - t_{\rm ref}}{P} \quad (1)$$

We then randomly select a value of τ between 0 and 1. Because we assume that the system is always observed at t_{ref}, this uniformly samples the time at which we observe the planet. We generate 41×41 pixel image cutouts, centered on the primary PSF.

After creating each model, we then inject it with noise. To estimate the level of noise in our combined Stage 3 data, we measure the noise distribution within an aperture around the white dwarf. Selecting an aperture with an inner radius of 10 pixels and an outer radius of 30 pixels, we find that the noise follows a Gaussian distribution with a standard deviation of 0.05 MJy/sr. We inject this noise into our binary models, scaled by a factor of $\sqrt{2}$ to account for our two sets of observations, and also add photon noise, modeled by a Poisson distribution.

We then attempt to recover 20,000 randomly generated binaries by fitting the same double PSF model described in §3.1. However, because we fit a single combined image at a time, we have just seven free parameters: the x and y coordinates of the target, the source and background flux, the offsets between the primary and secondary PSFs, and the difference in magnitude. We consider an injected companion recovered if the best-fit companion is within two pixels and one magnitude

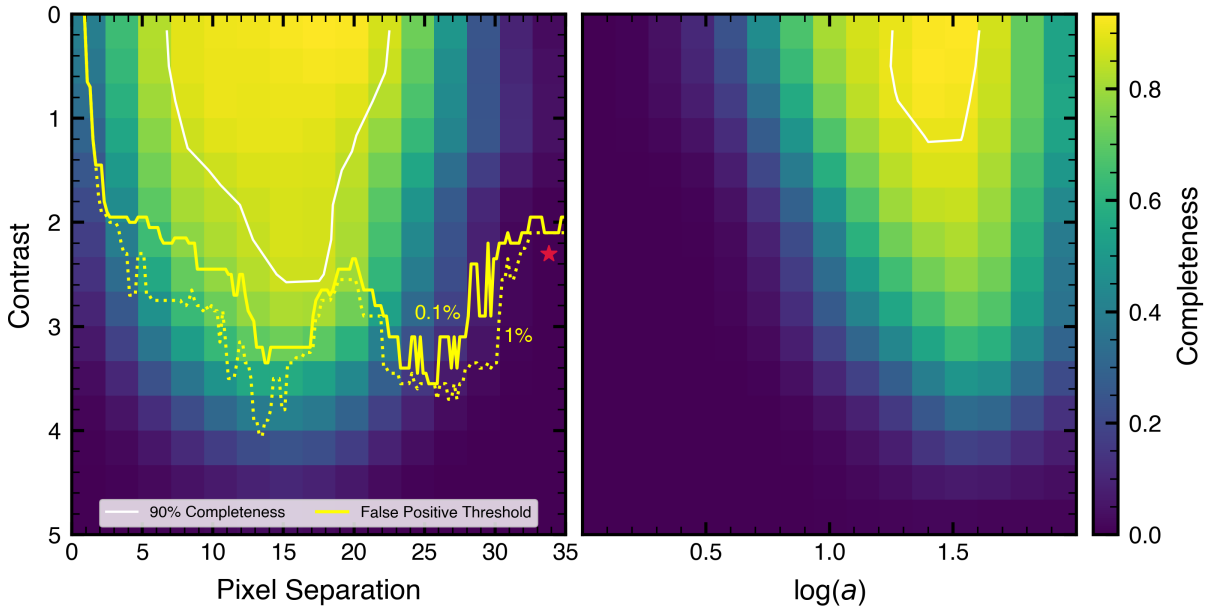


Figure 6. Completeness in the F1500W filter as a function of pixel separation (left) and semimajor axis (right). We randomly generate 20,000 synthetic planets with a range of contrasts and separations, as described in §4.1. We show both the 90% recovery threshold and the 0.1% false positive curve in white and yellow, respectively. We are able to reliably recover companions beyond ~5 au, and are primarily sensitive to companions with contrasts <2.6 mag. The best-fit companion, denoted by the red star, falls well outside the 90% completeness boundary and below the 1% false positive curve, suggesting it is not a real source.

of the known values. These values correspond approximately to where the distributions transition to become uniform, indicative of a non-detection. The resulting completion plot is shown in Figure 6. We find that we are able to recover companions down to separations of ~5 pixels, corresponding to a physical separation of ~14 au. Though it is difficult to detect faint companions this close in, we are able to recover wider separation companions with contrasts down to 2.6 mag.

To convert our contrast limits to mass, we use the approach outlined by Limbach et al. (2025). This ensures that our limits are consistent with the estimated mass of WD 1856 b. We first convert each contrast limit to a blackbody temperature, and then use evolutionary models (Fortney et al. 2007; Marley et al. 2021) to estimate the corresponding planet mass for a system age between 7.4–10 Gyr (Limbach et al. 2025; Vanderburg et al. 2020). We rule out >7.6–8.7 M_{Jup} companions at > 22 au. Our constraints are shown as a function of semi-major axis in Figure 7.

In addition to the completeness curve, we calculate the false positive probability as a function of separation and contrast. This defines the parameter space where our double PSF detections are reliable. As in De Furio et al. (2022) and Mehta et al. (2026), we generate 1,000 artificial single stars and attempt to fit them with the double PSF model. Each fit generates many samples with associated χ^2 values and corresponding χ^2 probabilities, $P(\chi^2)$. We then define a grid with pixel separations between 0 to 40 (with a bin size of 0.2) and contrasts between 0 and 7 mag (with a bin size of 0.05 mag). For each artificial source, we find the highest $P(\chi^2)$ within each bin and normalize the distribution to 1. We sum all 1,000 distributions, re-normalize the resulting grid to 1, and normalize each column at a given separation. This provides a cumulative false-positive percentage for each bin. We then calculate the 0.1% and 1% false positive probability curves, shown in Figure 6. We expect a double PSF fit to a single star to converge to a value above this curve only 0.01% of the time. We note that the curve falls below the 90% recovery line up to separations of 18.6 pixels, suggesting that any companion recovered above this limit is unlikely to be a false positive. However, our best-fit companion falls well outside the 90% completeness boundary and below the 1% false positive probability curve, suggesting that it is not a true companion.

4.2. *Proper Motion Limits*

To measure a source's proper motion, we must detect it in both epochs of imaging. We therefore measure the sensitivity of our proper motion search by estimating our sensitivity to sources in a single observation. We do this using Vortex Image Processing (`VIP`, Gomez Gonzalez et al. 2017; Christiaens et al. 2023), a Python package that uses injection recovery tests to calculate a contrast curve for a given image and sigma level. We first center the white dwarf using a positive two-dimensional Gaussian fit with the `ndimage-fourier` library, and perform PSF subtraction using the combined empirical PSF and STPSF described in §4.1 as a reference star. We then compute the 5σ contrast curve using the student-t cor-

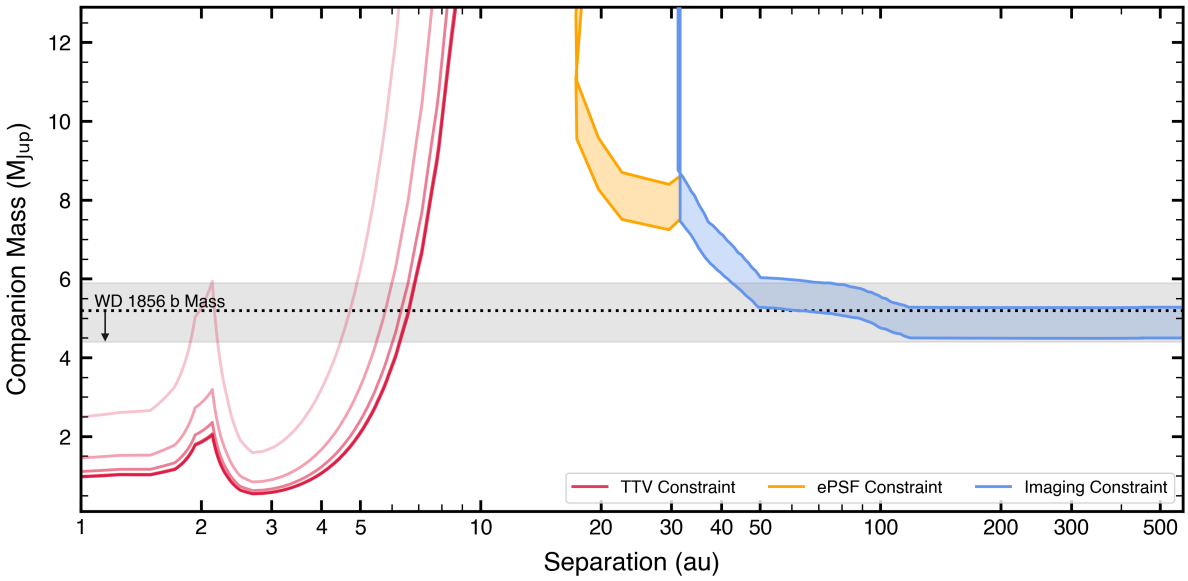


Figure 7. Combined constraints from TTVs and MIRI imaging. We include TTV constraints for companions at a range of inclinations (20, 40, 60, 80, and 90 degrees, with larger values denoted by increasing opacity), and imaging constraints for the system's expected age range (7.4−10 Gyr). We rule out planets $\gtrsim$1 M_{Jup} at close-in separations and planets $\gtrsim$5 M_{Jup} at wide separations. For comparison, we include an upper limit on the mass of WD 1856 b, with uncertainties denoted by the gray shaded region. Any less massive perturbing companion would likely have been ejected from the system.

rection to account for small-number statistics at small inner working angles (Mawet et al. 2014). We find that we reach the background-limited regime at ∼2″. To avoid contamination from nearby sources, we calculate the average sensitivity between 2−3″, and assume that this is representative of the sensitivity throughout the background-limited regime. To convert our constraints from units of contrast to mass, we use the conversion described in §4.1.

However, `VIP` only calculates the contrast curve in terms of a planet's observed separation, rather than its semimajor axis. To account for this, we use an adapted version of the Exoplanet Detection Map Calculator (`Exo-DMC`) Python package (Bonavita 2020) to measure the detection probability of a synthetic planet population. Because the default package only allows for a uniform or Gaussian eccentricity distribution, inconsistent with the eccentricity distribution we assume for our target injection in §4.1, we modify the code to use the Beta distribution described by Nagpal et al. (2023) (α = 0.7, β = 2.3). We take the 90% completeness curve from the resulting probability distribution as our sensitivity limit, shown in Figure 7. We find that we are able to rule out companions >5.3−6.1 M_{Jup} at separations of 50−110 au and >4.5−5.3 M_{Jup} at separations >110 au, where the provided ranges correspond to lower and upper limits on the system's age.

4.3. *TTV Limits*

We then extend existing TTV constraints on the presence of any additional planets in the system. Using the combined datasets from Kubiak et al. (2023) and Gendreau-Distler et al. (2025), in addition to NIRSpec PRISM and qCMOS observations, we calculate a best-fit TTV model. Following Kubiak et al. (2023), we use `ttvfast` (Deck et al. 2014) to predict the transit times, and add an additional perturbation from the Roemer delay, which can dominate in systems with a wide companion orbiting a close inner pair. We assume that WD 1856 has an inclination of 90 degrees and that both WD 1856 b and any outer companion have an eccentricity of zero. For a given set of orbital parameters, we then calculate the likelihood of the resulting model. Using this likelihood function, we perform a grid search across varying values of a companion's mass and period, calculating the likelihood for a total of 300 mass values and 100 period values. We repeat this search for a variety of companion inclinations (20, 40, 60, 80, and 90 degrees). For more details on the TTV model, likelihood function, and grid search, see Kubiak et al. (2023). For a list of all transit observations (excluding the NIRSpec and qCMOS observations), see Table 4 of Gendreau-Distler et al. (2025).

We then calculate the likelihood ratio between a one-planet model and a two-planet model (taken to be the model with the highest likelihood within each grid point). We consider a parameter space ruled out if the likelihood ratio is greater than 10^3, as this provides strong evidence that the one-planet model is favored. We show the resulting constraints in Figure 7. For an inclination of 90 degrees, we rule out planets > 2.3 M_{Jup} within 5 au, and planets more massive than WD 1856 b within 6.6 au. For an inclination of 20 degrees, we are less sensitive; we rule out planets > 6.2 M_{Jup} within 5 au, and planets more massive than WD 1856 b within 4.7 au.

5. DISCUSSION

5.1. *Implications for the Dynamical History of WD 1856 b*

A promising explanation for WD 1856 b's close-in orbit is that it underwent a planet-planet interaction before circularizing via high-eccentricity migration (Maldonado et al. 2021; O'Connor et al. 2022). Assuming that the planet is currently on a circularized orbit and that the planet must have been at an initial orbital separation $\geq$ 2 au to survive its host star's transition off the main sequence, we use conservation of angular momentum to estimate that the planet must have been excited to an eccentricity $e \geq 0.995$ to reach its present location. One potential line of evidence for a high-eccentricity migration scenario is the planet's orbit; planets that undergo high-eccentricity migration are expected to pile-up around $\gtrsim 2a_{\rm Roche}$ (Rasio & Ford 1996; Matsumura et al. 2010), consistent with WD 1856 b's semimajor axis of 4.4±0.9 $a_{\rm Roche}$.

However, in this work we rule out high-mass companions throughout much of the parameter space. This suggests that, if WD 1856 b did undergo a planet-planet interaction, its companion is below our detection limits or was ejected from the system. Because ejection is a fairly common outcome of planet-planet scattering, with the less massive companion nearly always being the one ejected (e.g., Moorhead & Adams 2005; Raymond et al. 2010; Marzari 2025), we expect any low-mass perturber to have been ejected from the WD 1856 system. There is therefore limited remaining parameter space where we might find a perturbing companion. Some simulations suggest that the joint probability of a planet undergoing high-eccentricity migration while also ejecting all companions is small ($\sim$ 0.1%, Rasio & Ford 1996; Anderson et al. 2020). Therefore, if we are able to conclusively rule out the presence of any planet of similar or larger mass than WD 1856 b, we may be able to conclude that WD 1856 b did not migrate inward via planet-planet interactions.

We note that the mass estimate for WD 1856 b in Table 1 assumes that insolation contributes negligibly to its thermal evolution. However, the permitted mass range extends to 0.84−5.9 $M_{\rm Jup}$ if insolation has non-negligible contributions (Limbach et al. 2025) and we adopt a lower mass limit from Xu et al. (2021). Additionally, the planet may appear brighter if it has a significant day-night temperature difference. Because we are observing the planet at an orbital phase when more of the dayside is visible, this would lead to a larger measured effective temperature. See Limbach et al. (2025) for a detailed discussion on heat redistribution in WD 1856 b's atmosphere. This effect would be enhanced if the planet has a low albedo, which is unlikely for such a cold object. Despite this uncertainty regarding WD 1856 b's mass, even a 5.2 $M_{\rm Jup}$ planet could possibly be perturbed by a companion down to roughly 1 $M_{\rm Jup}$. We discuss how such a companion could be found in future searches in §5.2.

It is unclear whether multiple giant planets could have formed in this system. The progenitor of WD 1856 was likely a F or A-type star. While giant planets may be relatively common around these massive stars, if WD 1856 formed ~10 Gya, the system may have had a lower metallicity than F and A-type stars today. Given the known correlation between stellar metallicity and the presence of giant planets (Fischer & Valenti 2005; Johnson et al. 2010; Mortier et al. 2013, e.g.,), we would expect a lower progenitor metallicity to correspond to fewer giant planets within the system. However, we note that the stellar age-metallicity relationship has significant scatter, and may be relatively flat in the solar neighborhood (e.g., Haywood et al. 2013; Bergemann et al. 2014; Delgado Mena et al. 2019; Rebassa-Mansergas et al. 2021). It therefore remains possible that multiple giant planets formed around WD 1856's progenitor star.

5.2. *Future Searches for a Companion*

We expect a 5 $M_{\rm Jup}$ planet to be ~24 times brighter in the F1500W filter when compared to a 1 $M_{\rm Jup}$ perturber, and ~16 times brighter in the F1800W filter. Though we note that brightness is also strongly dependent on the planet's metallicity, this suggests that our imaging program would have to be greatly extended to fully rule out these lower mass companions, requiring over 70 hours of additional imaging.

TTVs and astrometry provide alternative methods for detecting lower mass companions. With the current TTV baseline of 5.8 years, we are only sensitive to WD 1856 b-mass planets out to 6.6 au. However, as we increase this baseline, we will also increase our sensitivity to wide-orbit companions. Astrometry may also rule out some of the parameter space. With the upcoming release of Gaia DR4 (Gaia Collaboration et al. 2016b), we expect to be sensitive to planets down to nearly 2 $M_{\rm Jup}$ at separations of ~4 au. Additionally, we expect to be sensitive to planets at least 10 $M_{\rm Jup}$ throughout the 1−10 au region. These predicted Gaia DR4 constraints are only more sensitive than existing TTV constraints for planets beyond ~6 au. Unlike imaging, improved constraints from both TTVs and astrometry will be independent of the planet's atmospheric composition and structure. Therefore, if there is a faint, low metallicity companion, it may be easier to detect it with these alternative methods.

6. CONCLUSION

In this paper, we place constraints on the presence of any outer companions to WD 1856 b. The main conclusions are summarized as follows:

- Using JWST MIRI imaging, we rule out >7.6−8.7 $M_{\rm Jup}$ planets at >22 au, >5.3−6.1 $M_{\rm Jup}$ planets at 50−110 au, and >4.5−5.3 $M_{\rm Jup}$ planets between >110 au and WD 1856's Hill sphere radius. The provided ranges correspond to lower and upper limits on the system's age (estimated to be 7.4−10 Gyr).

- Using updated TTV constraints, we also rule out planets > 2.3 $M_{\rm Jup}$ at close-in (<5 au) separations. This limit increases for higher inclination planets.

- Combined, our constraints rule out high-mass companions across much of the searched parameter space. If further observations rule out the remaining region for a planet more massive than WD 1856 b, then it is likely

that any perturbing companion was ejected from the system.

WD 1856 b offers rare insight into the forces that shape post-main-sequence planetary systems. Better understanding its dynamical history not only informs our understanding of this unique system, but also provides a glimpse into the future of the thousands of known exoplanet systems.

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Software: matplotlib (Hunter 2007), numpy (Harris et al. 2020), Astroalign (Beroiz et al. 2020), Photutils (Bradley et al. 2016), MADYS (Squicciarini & Bonavita 2022)

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